1D HEAT EXCHANGER SOLUTION- HOLMAN

Daniel

$NTU = \frac{UA}{C_{min}}$

$C = C_{min}/C_{max}$

COUNTER FLOW DOUBLE PIPE:

$\epsilon = \frac{1-\exp[-NTU(1-C)]}{1-C\cdot\exp[-NTU(1-C)]}$

WATER-WATER C=1:

$\epsilon = \frac{NTU}{NTU+1}$

$q_{max} = (\dot{m}c)_{min}(T_{h,in} - T_{c,in})$

$q_{actual} = (\dot{m}c)_{min}(T_{minfluid,in} - T_{minfluid,out})$

MIN FLUID IS THE HOT FLUID:

$T_{h,out} = T_{h,in} - \epsilon(T_{h,in} - T_{c,in})$

MIN FLUID IS THE COLD FLUID:

$T_{c,out} = T_{c,in} + \epsilon(T_{h,in} - T_{c,in})$

ALSO:

$NTU = \frac{\epsilon}{1-\epsilon}$

$NTU = \frac{1}{C-1}ln(\frac{\epsilon-1}{C\epsilon-1)})$